



Healthcare Analysis

Check out the
project online!



The Outliers Team

- **Abdulrahman Waheed**
- **Hoda Elkady**
- **Abdulrahman Hisham**
- **Mohammad Abdulhakim**

Overview

The goal behind this project is to provide a comprehensive analysis of healthcare datasets, giving an example of the correct approach to tackling this type of dataset.

The project will go through four phases: Cleaning, Modeling, Visualization and then Revision.

In the Cleaning phase, we will clean the dataset to make it ready for analysis.

In the Modeling phase, the cleaned dataset will be split into entity tables and the relationships between all of these tables will be determined, then finally we will create DAX measures for data insights.

In the Visualization phase, the previously calculated DAX measures will be used to create a beautiful report on the dataset

Finally, in the Revision phase, we will go over any ultimate changes that need to be made to the dataset

Dataset



The dataset consists of 18 tables covering different aspects of clinical data

allergies	careplans	claims
claims_transactions	conditions	devices
encounters	medications	observations
imaging_studies	immunizations	organizations
patients	payers	payer_transitions
procedures	supplies	providers

SyntheticMass is a synthetic healthcare dataset modeling fictional residents of Massachusetts, complete with realistic, but artificial health records generated by the Synthea tool. It provides a privacy-safe sandbox free of real PHI or PII for testing health IT applications, data analysis, and research.

Phase 1

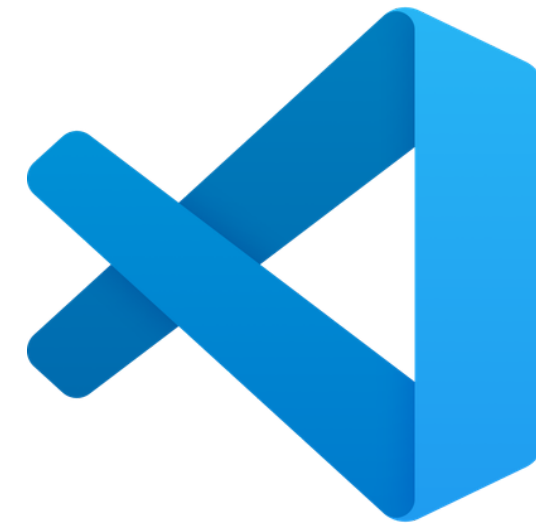
Data Cleaning

The tools and methods used to clean the dataset

Data Cleaning

The cleaning of the data was entirely done using Python libraries, such as pandas and numpy

IDE: Visual Studio Code



Language: Python



pythonTM

Data Cleaning Common Steps

Most columns had a similar issues within them. This is how we dealt with these issues:

- STOP column is often empty so it is filled with the current date instead. (In the used data nulls are replaced with May 1st 2026, the date when the cleaning process originally took place)
- Columns that are duplicated or irrelevant to the analysis were removed.
- Some columns were transformed into a category type for better optimization.
- Replace the null values with proper or stand-in equivalent values.
- Columns that had an incorrect type were corrected

Phase 2

Data Modeling

EDA, Relational analysis and measures calculation

Data Modeling

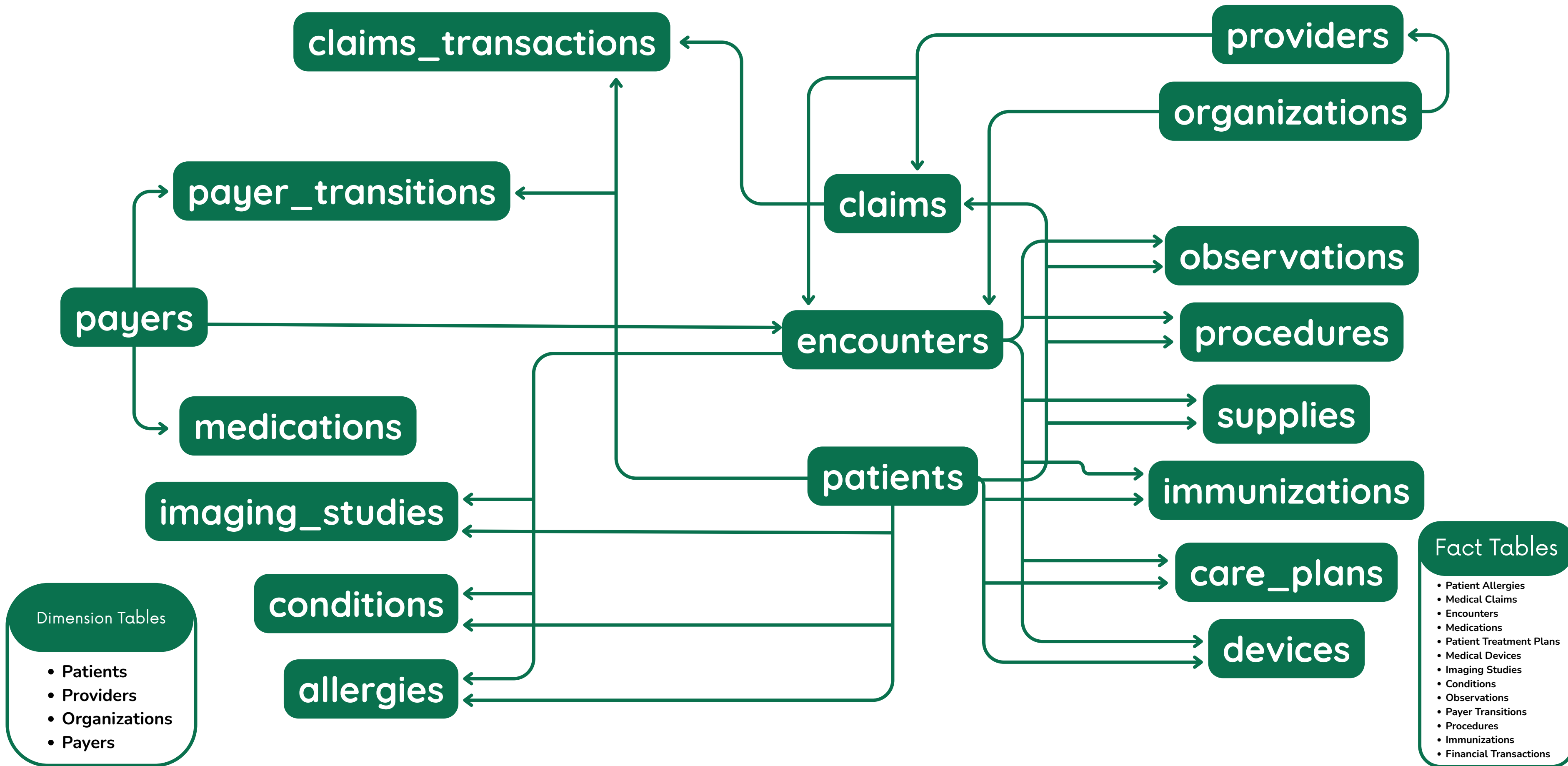
The data model building and measures calculations were made in Microsoft Power BI

Modeling Tool: Microsoft Power BI



Data Modeling

ERD



Data Modeling KPIs and Measures

Patients and Demographics

- Total Patients
- Average Income
- Total Patient Cities
- Unique Patients Visited
- Unique Patients Observed
- Total Healthcare Expenses
- Average Health Coverage
- Unique Patients Imaged
- Unique Patients with Claims
- Unique Patients with Conditions
- Total Patients with Transitions

Clinical & Diagnostics

- | | | |
|--|--|---|
| <ul style="list-style-type: none">• Total Encounters• Avg Cost Per Encounter• Avg Encounter Duration in Minutes• Emergency Encounters• Insurance Coverage Rate• Total Base Cost• Total Insurance Covered• Total Out-of-Pocket Cost• Total Conditions• Active Conditions• Total Unique Diseases• Avg Conditions Per Encounter• Avg Conditions Per Patient | <ul style="list-style-type: none">• Total Encounters with Conditions• Total Procedures• Total Procedure Cost• Average Cost per Procedure• Avg Procedures per Patient• Total Unique Patients• Procedures Cost LY• Total Observations Count• Vital Signs Observations• Laboratory Observations• Laboratory Observations Percentage• Avg Systolic New• Avg Diastolic New• Average Body Weight• Total Allergies• Unique Patients with Allergies | <ul style="list-style-type: none">• Allergy Density• Average BMI New• Total Imaging Studies• Avg Studies Per Patient• CT Scans Count• CT Scan Rate• Thoracic Imaging Count• Modality Contribution• Severe Allergies Count• Severe Allergies Rate• Moderate Allergies Count• Current Month Allergies• Allergies Previous Month |
|--|--|---|

Data Modeling KPIs and Measures

Financial & Costs

- Number of Transactions
- Transactions Sum
- Transfers Sum
- Total Immunization Base Cost
- Average Immunization Base Cost
- Total Dispenses
- Total Medications Cost
- Average Cost Per Medication
- Total Payer Coverage
- Medication Insurance Coverage Rate
- Out-of-Pocket Rate
- Total Claims Count
- Insured Claims Count
- Insurance Coverage Claims Rate
- Claims Distribution
- Total Supplies
- Total Supply Records

Providers & Organizations

- Total Providers
- Total Provider Encounters
- Total Cities Covered
- Male Encounters
- Female Encounters
- Average Provider Encounters
- Total Organizations
- Total Cities
- Total Utilization
- Avg Utilization
- Max Utilization
- Unique Providers

Payers & Coverage

- Total Revenue
- Total Payers Count
- Total Amount Covered
- Total Amount Uncovered
- Total Uncovered Expenses
- Total Covered Encounters
- Total Covered Medications
- Total Unique Customers
- Total Transitions
- Active Transitions
- Self Paid Transitions
- Transitions This Year
- Patients with Multiple Transitions
- Avg Coverage Duration (Days)
- Uninsured Rate

Phase 3

Visualization

Dashboard design and Finalization

Visualization

The Visualization process is also done using Microsoft Power BI

Visualization Tool : Microsoft Power BI



116

Total Patients

6K

Total Encounters

\$31.74M

Transactions Sum

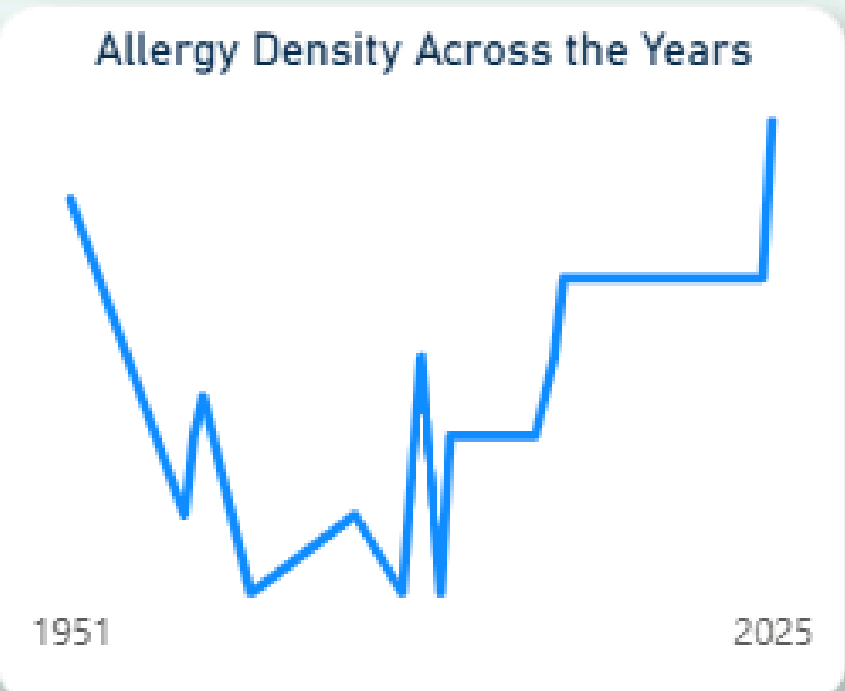
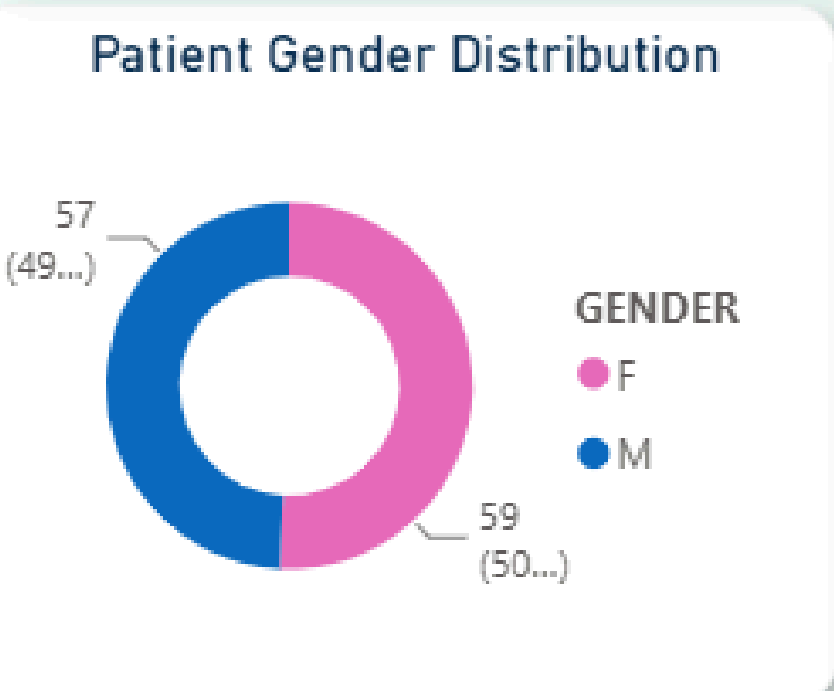
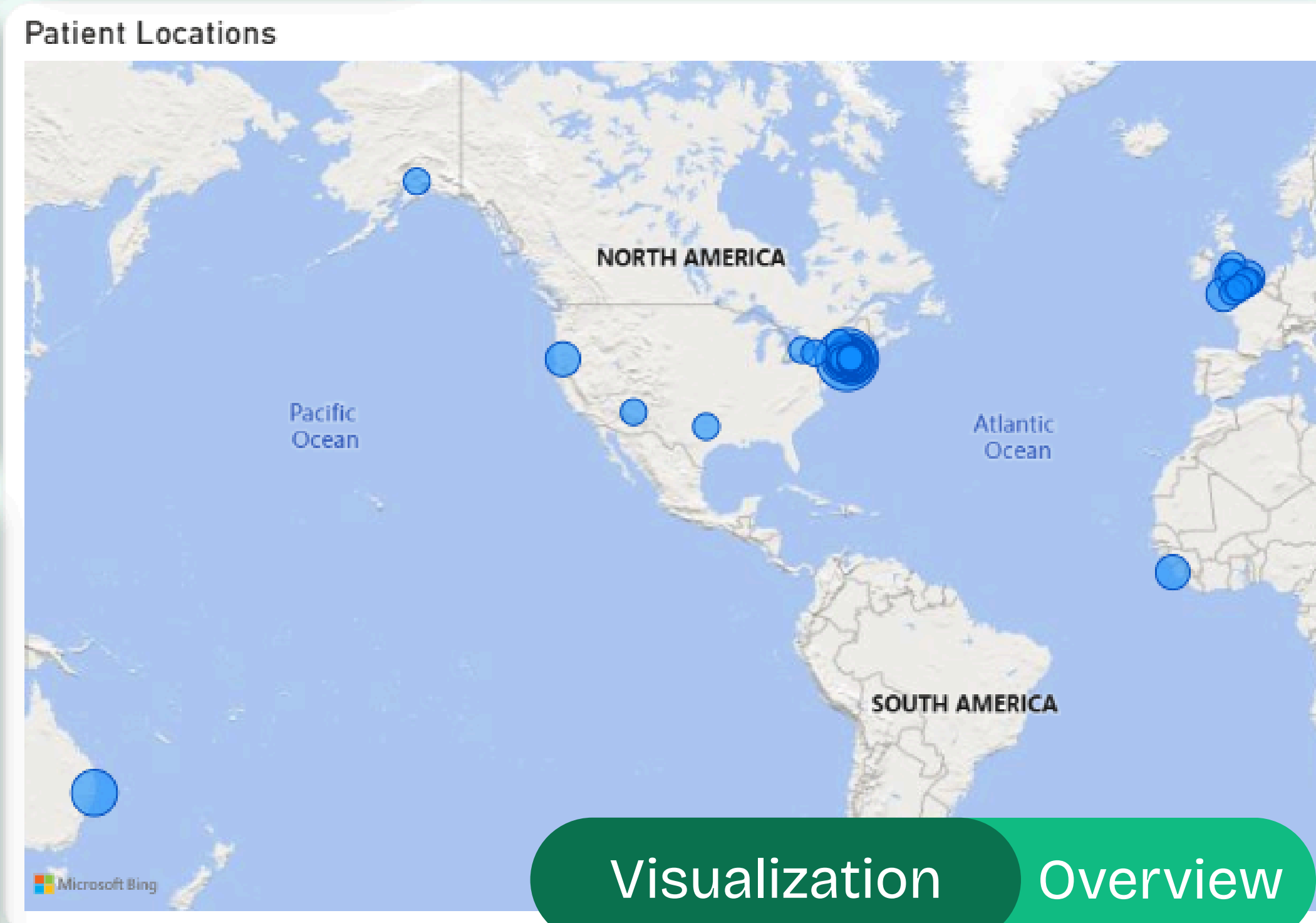
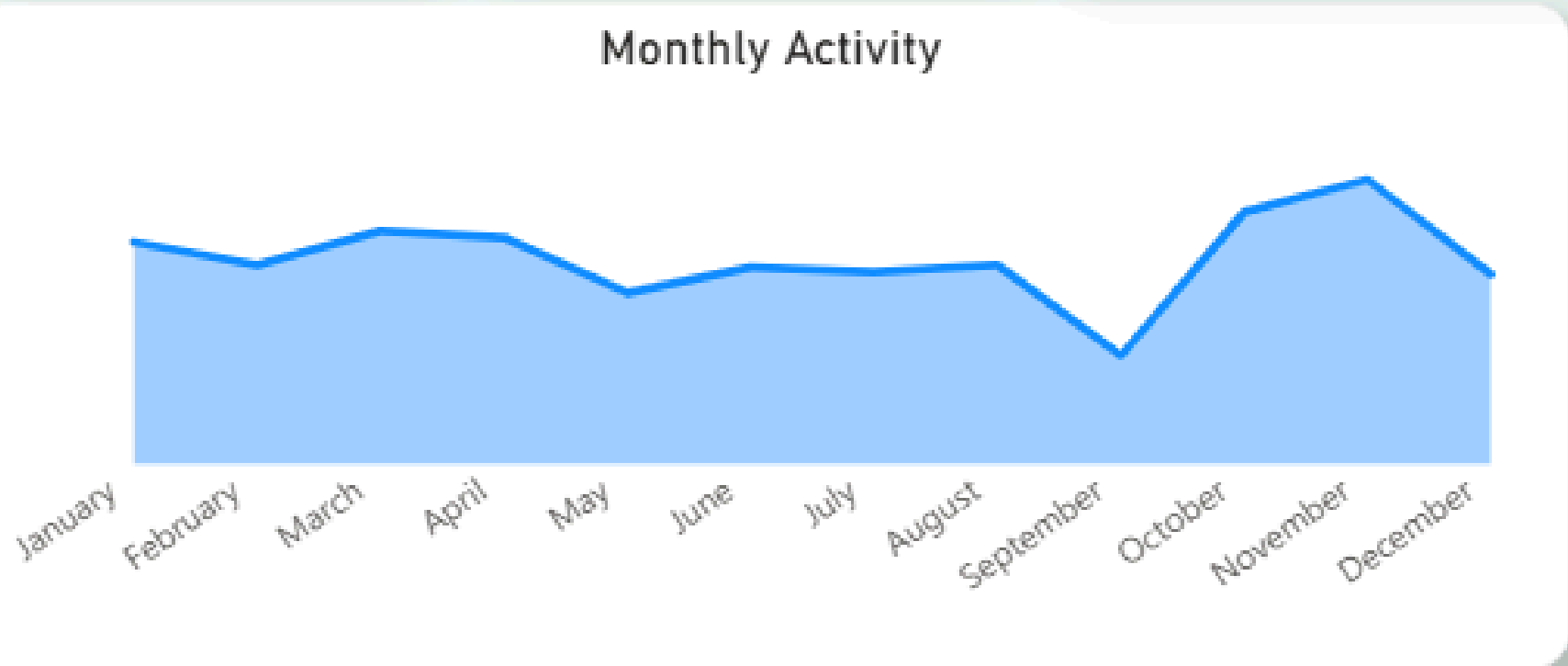
\$4.70M

Transfers Sum

Encounter Date

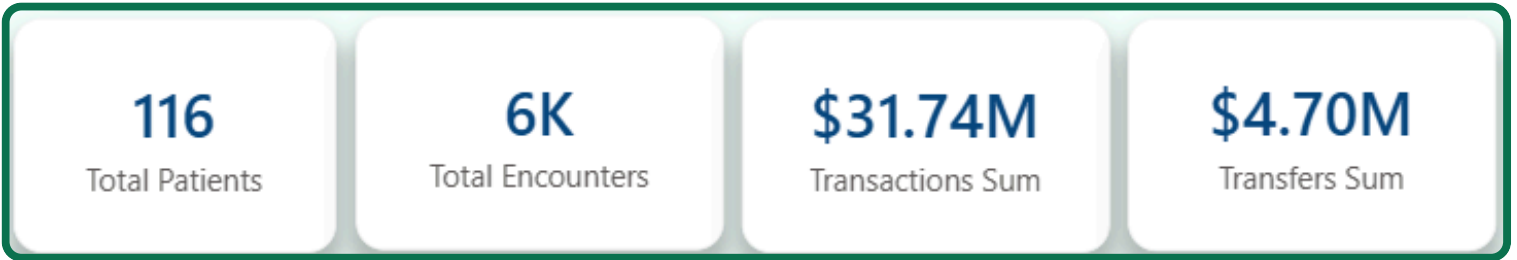
1/15/1944

3/5/2026

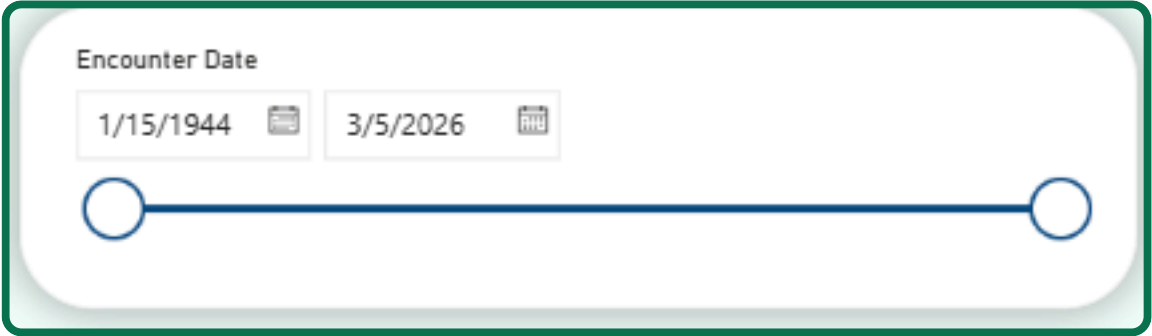


Visualization Overview

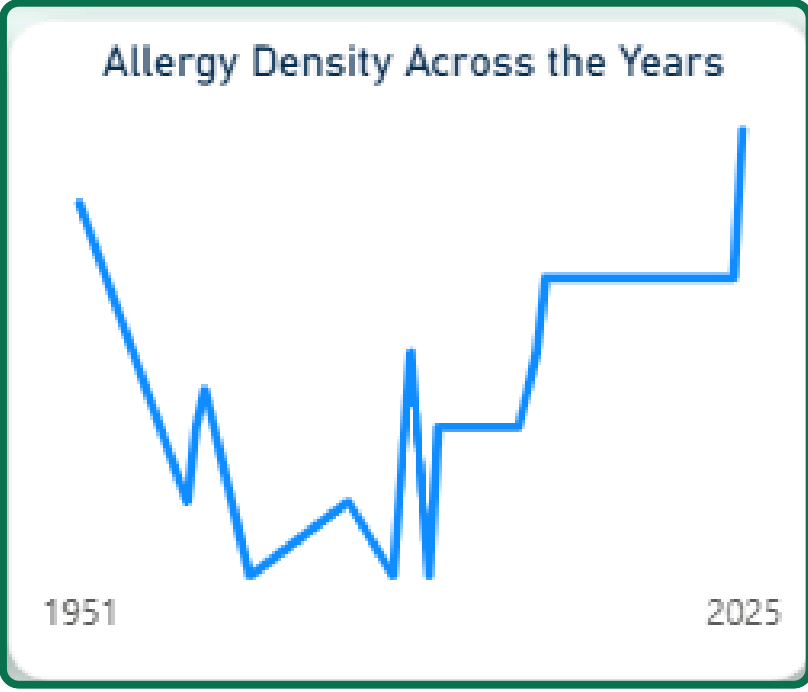
Monthly activity
based on
encounters



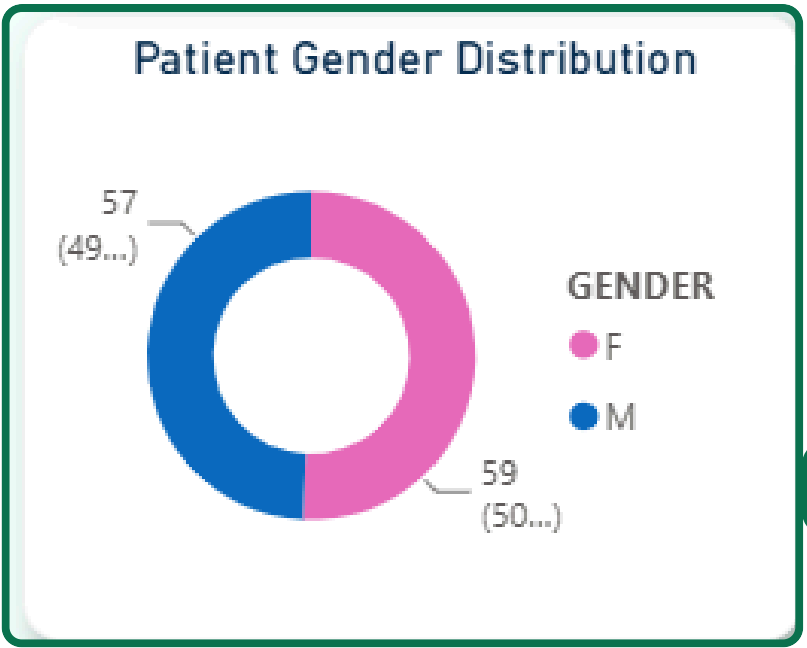
Basic measures



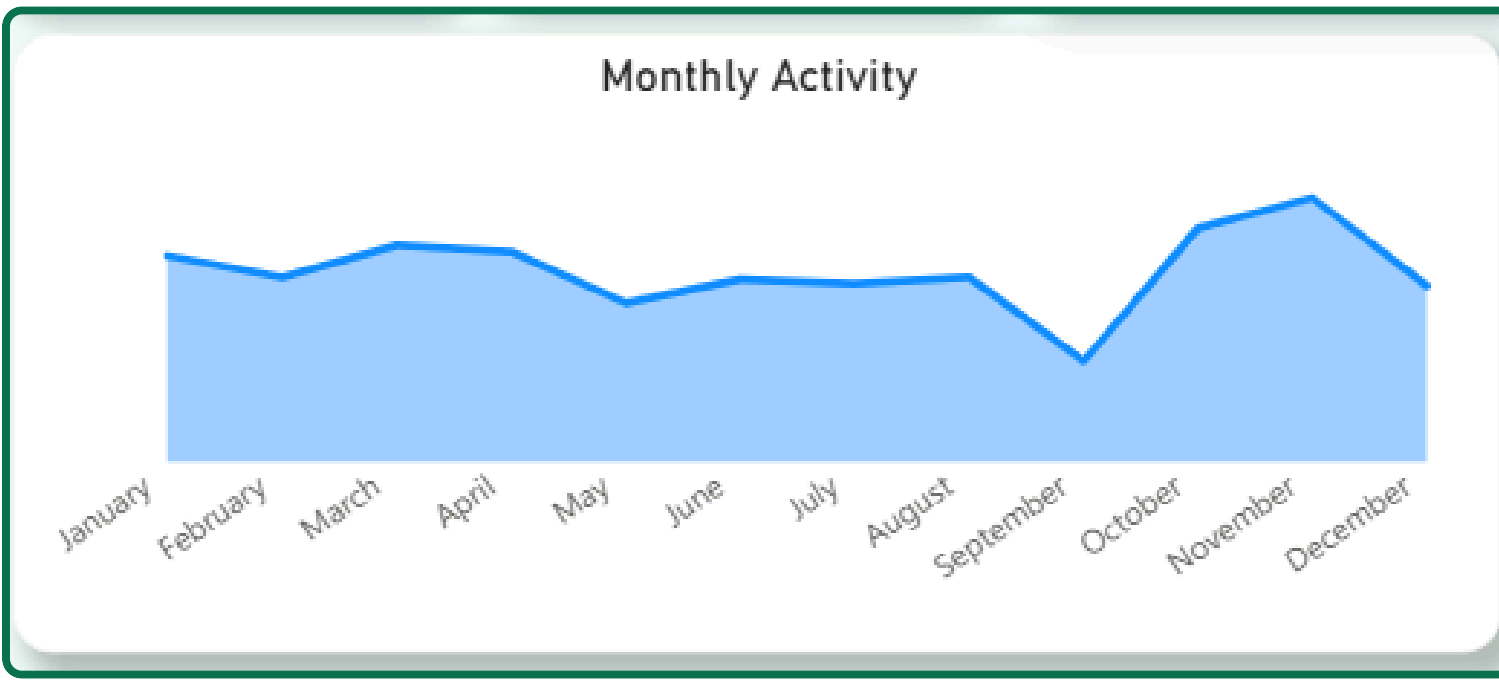
A slicer that changes time based
on encounter start date



A line chart showing the
change in allergies per
patient over the years

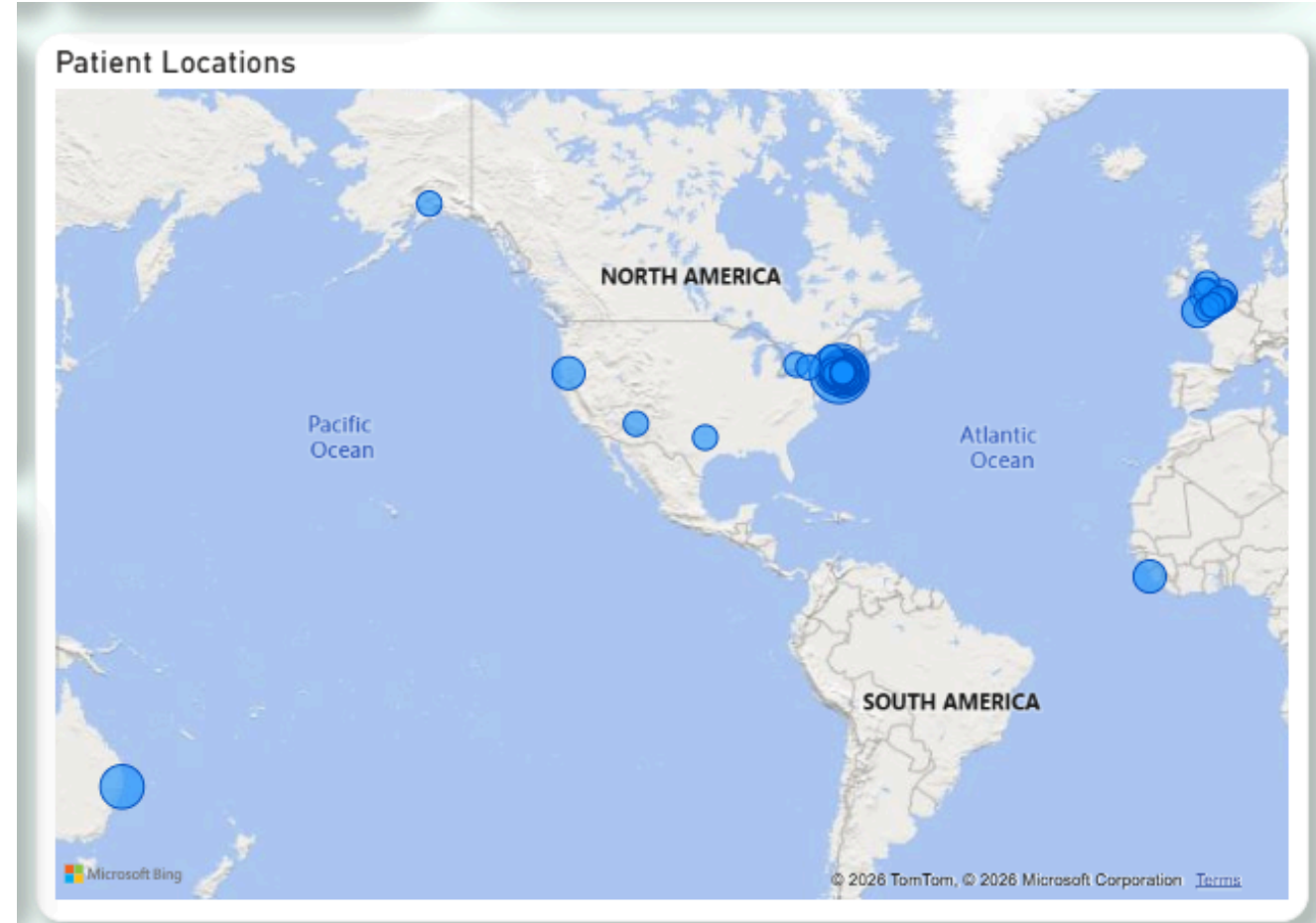


A donut chart showing the
distribution of patients based
on gender

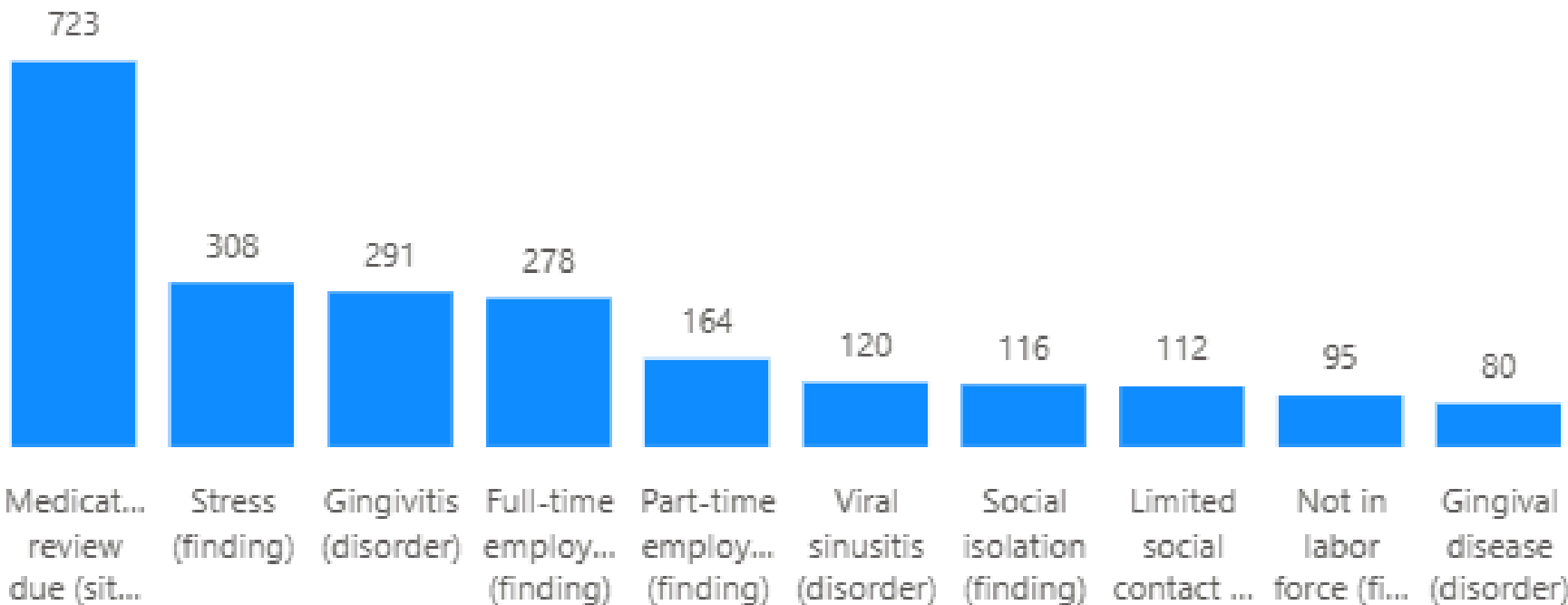


Navigation Bar

Patient distribution across the world



Most Common Conditions



FROMDATE

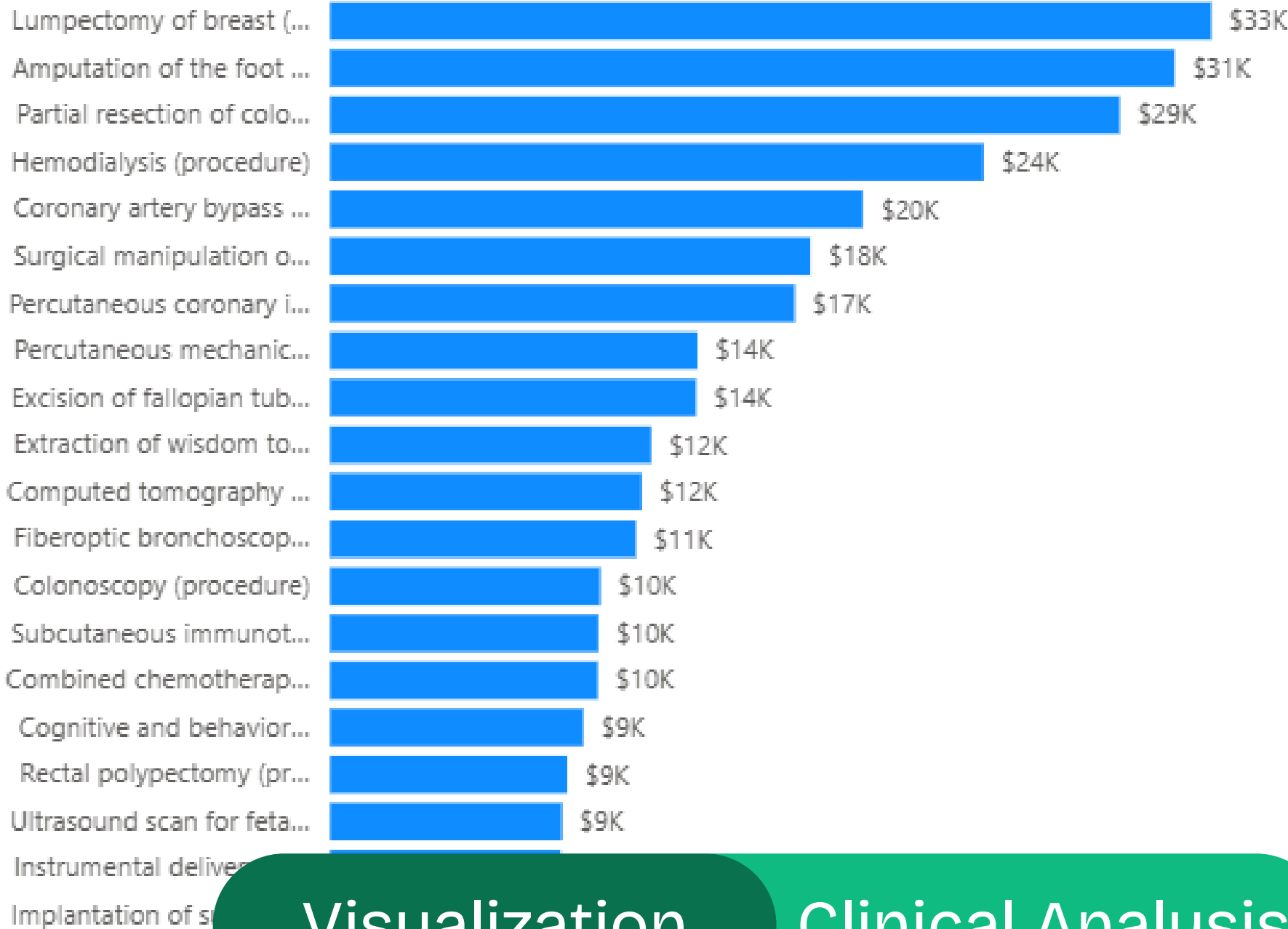
1/15/1944



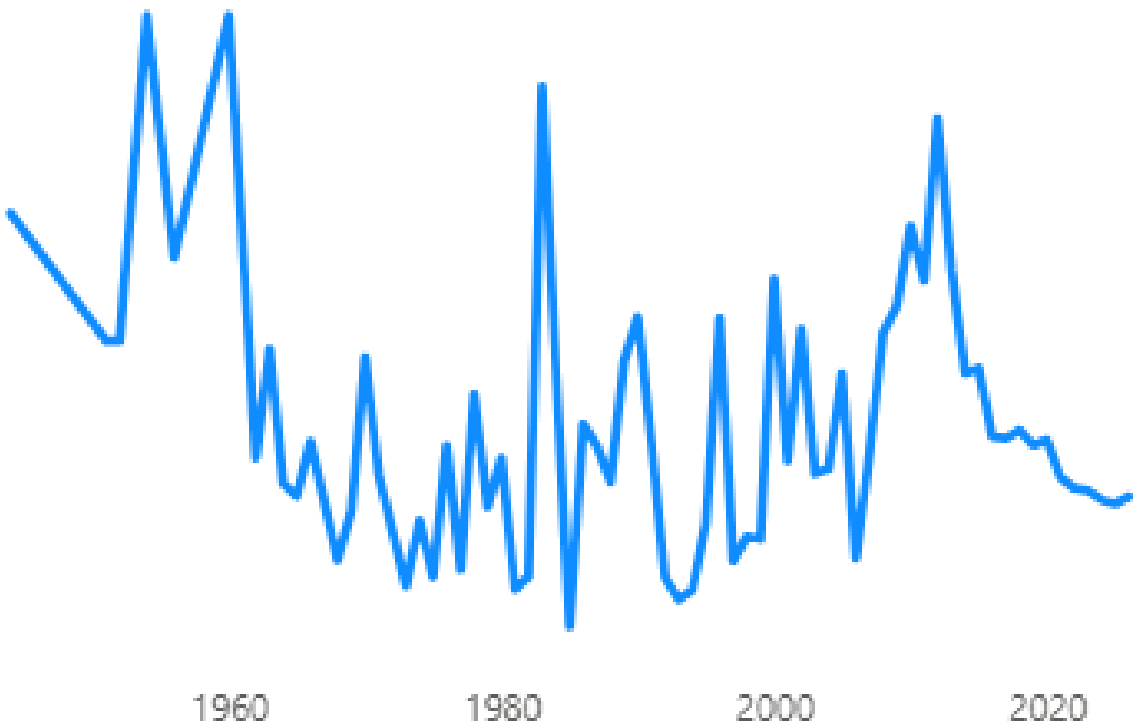
3/5/2026



Top Procedure by Average Cost



Average Healthcare Expenses Timeline



27.18

Average BMI

0.15

Critical Cases Rate

87K

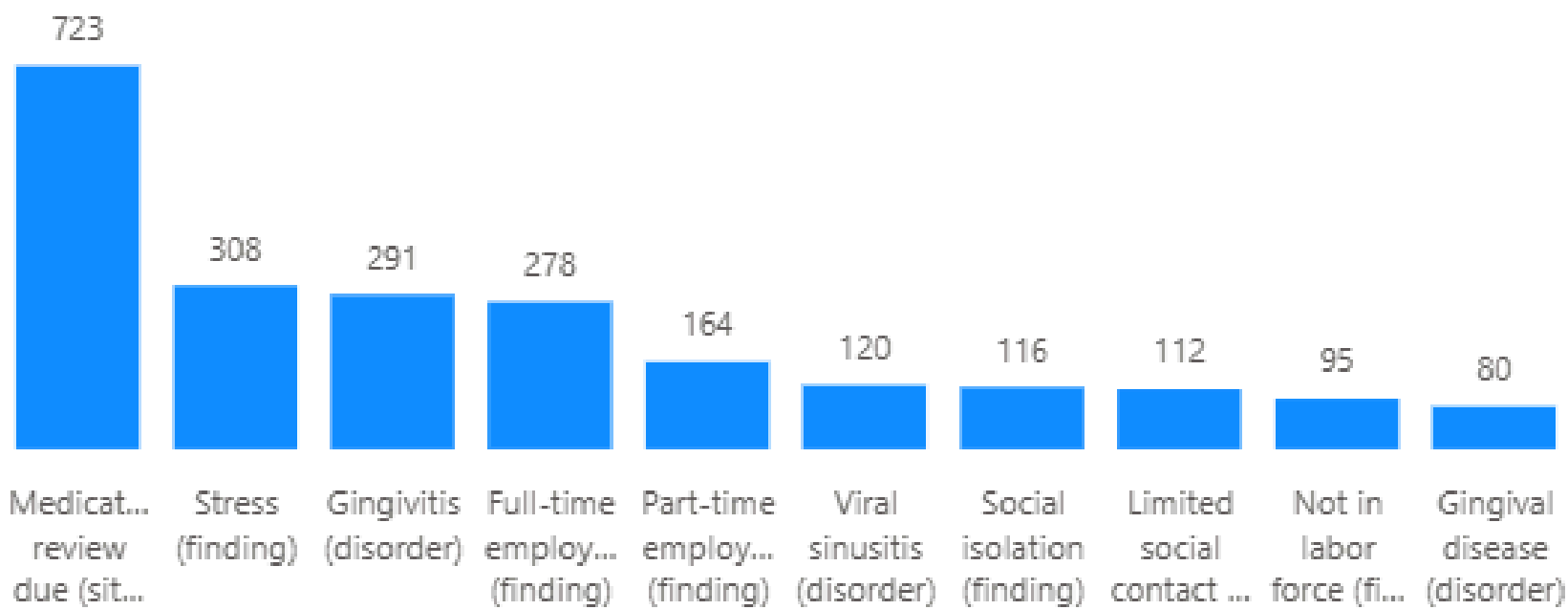
Total Observations

Visualization

Clinical Analysis

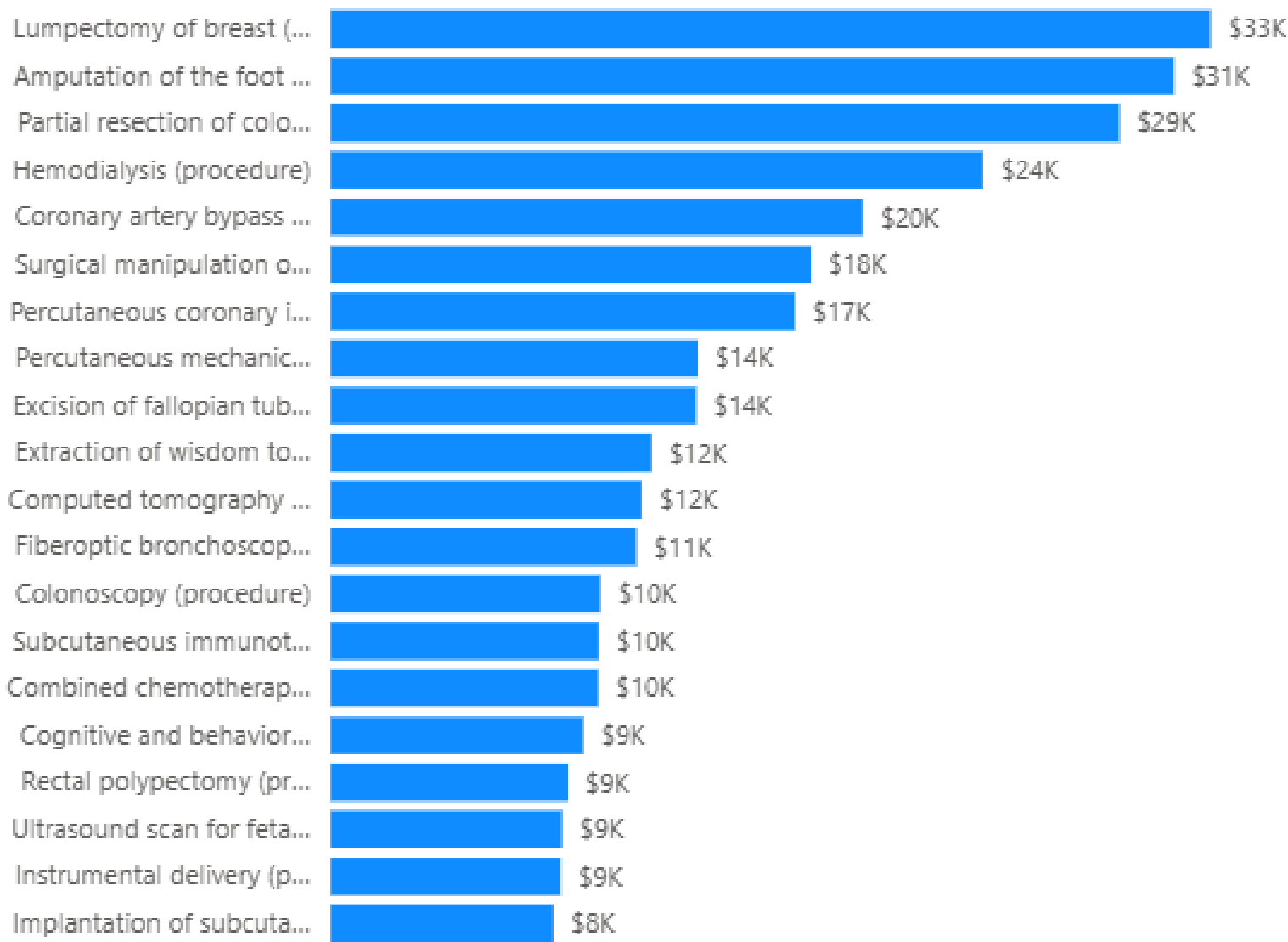
Visualization Clinical Analysis

Most Common Conditions



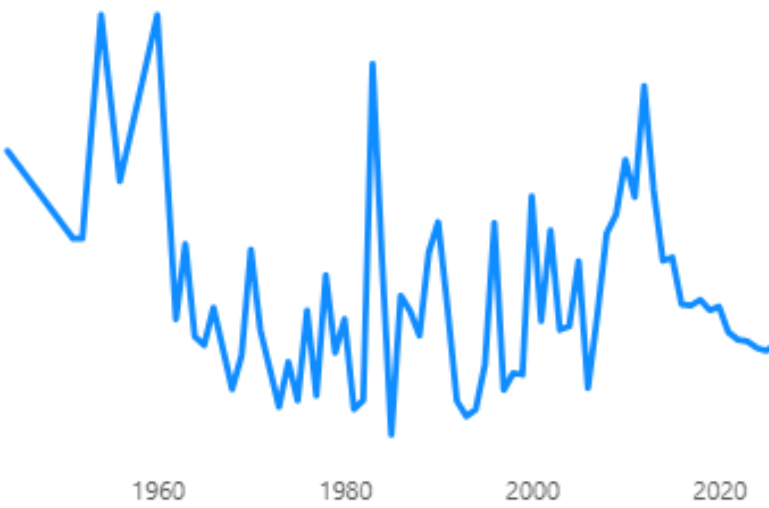
Clustered Column Chart of the Top 10 common conditions among patients

Top Procedure by Average Cost



Clustered Bar Chart of the Top 20 Procedures by their average cost

Average Healthcare Expenses Timeline



27.18

Average BMI

0.15

Critical Cases Rate

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Total Observations

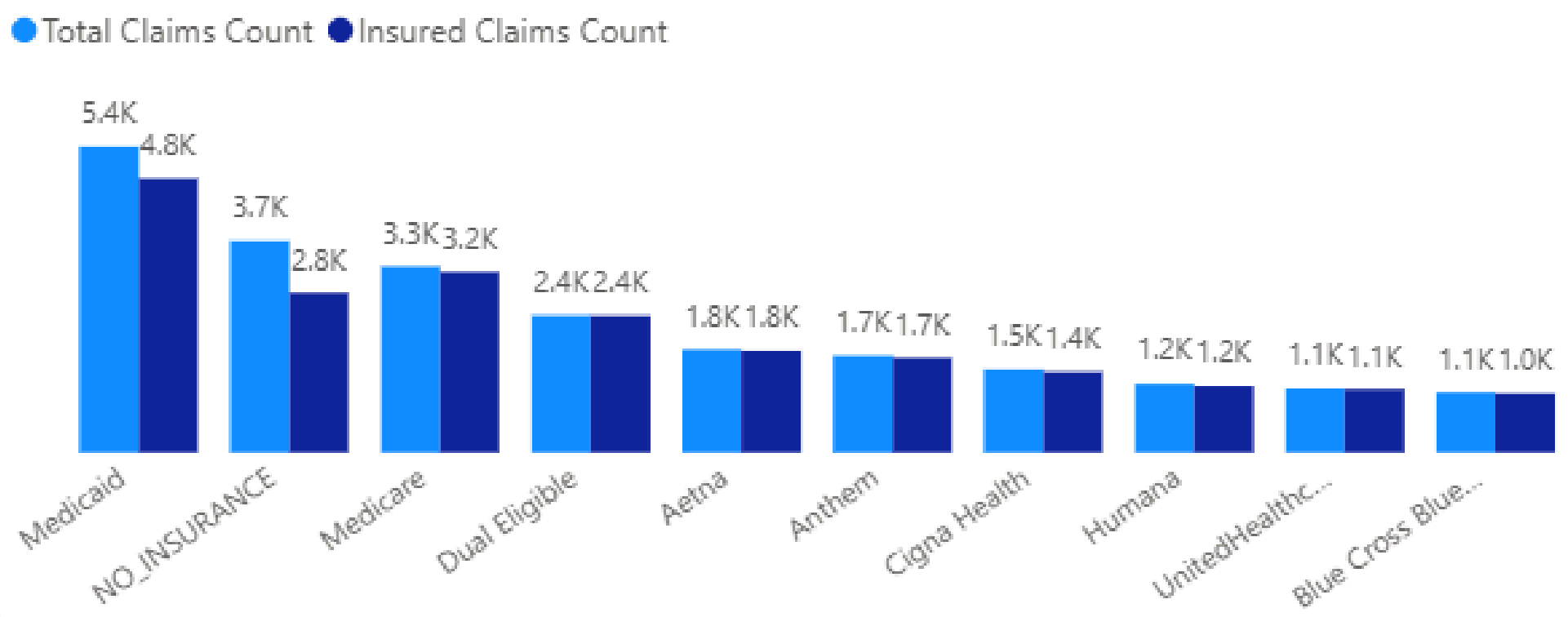
A Line Chart showcasing The evolution of average healthcare expenses across the years alongside some extra major measures

Slicer changing time based on procedure start date

Procedure Date



Distribution of Payers by Insured vs Uninsured Claims



Payer Coverage Information

NAME	Total Transitions	Uninsured Rate	Avg Coverage Duration (Days)
Medicare	1904	0.23	365.02
NO_INSURANCE	1694	0.56	365.01
Medicaid	1418	0.15	364.43
Cigna Health	1209	0.22	364.01
Aetna	1088	0.17	364.20
Humana	1057	0.25	365.21
Anthem	768	0.29	363.80
Blue Cross Blue Shield	630	0.25	365.21
UnitedHealthcare	359	0.33	365.23
Dual Eligible	268	0.00	363.84
Total	4810	0.25	364.84

5K

Total Transitions

0.25

Uninsured Rate

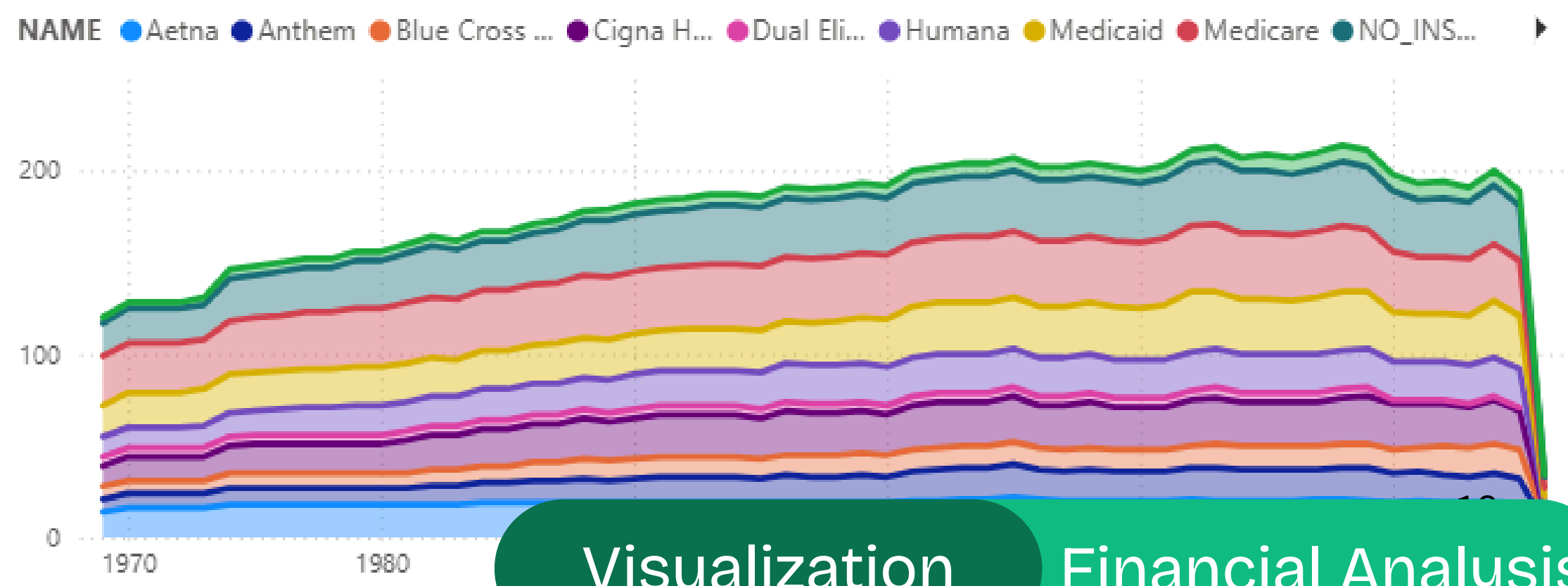
\$1.34K

Average Procedure Cost

\$27.19M

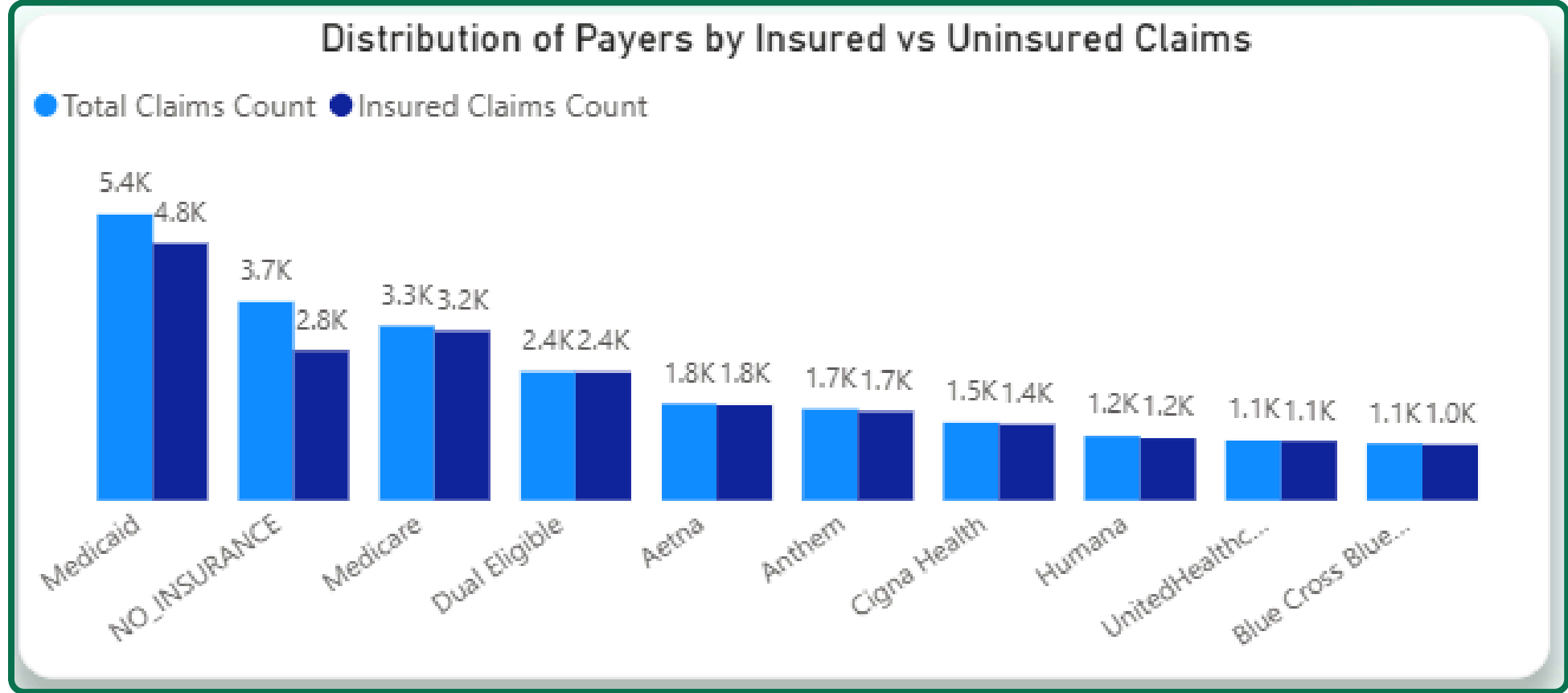
Total Procedure Cost

Trends by Payer



Visualization

Financial Analysis



Clustered column chart showcasing the difference between insured and uninsured claims for every payer

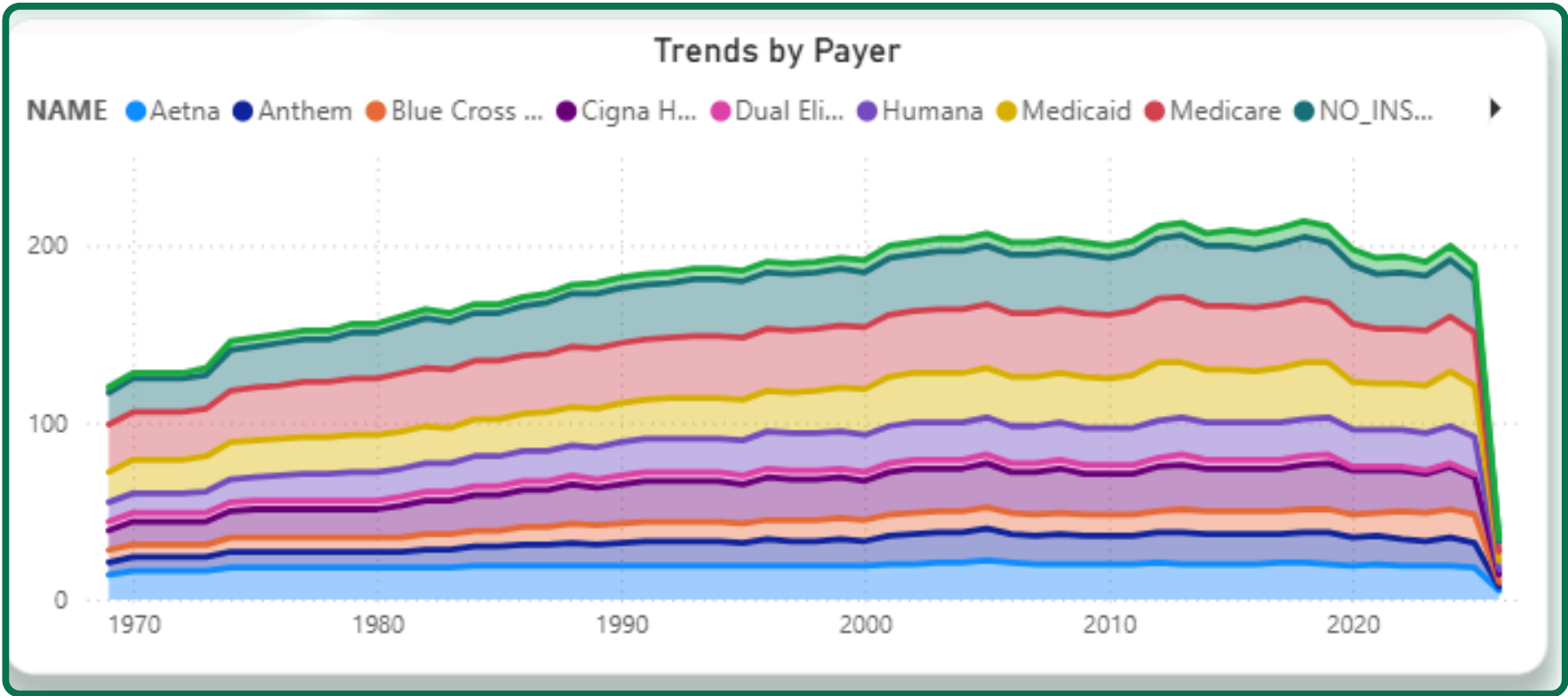


Stacked area chart showing number of transitions across different payers

Financial measures

Payer Coverage Information			
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A table showing every payer's total transitions, uninsured rate and average coverage duration



- **Lupectomy of breast and foot amputations are the most common procedures.**
- **Average healthcare expenses keep going up and down, but have decreased over all between 1960 and 2020.**
- **Most patients come from the United States and England.**
- **Activity is at its peak during November and at its lowest during September, which is interesting considering that the difference between them is not that large.**

Insights

- Allergy density saw a huge drop from 6.00 to 2.00 between 1951 and 1963, followed by a period of constant spiking and dropping until reaching a period of stability between 2003 and 2024 followed by an all-time high density of 7.00 during 2025
- Female patients are generally more common than male patients (52.21% against 47.79%)
- UnitedHealthCare has the highest average coverage duration followed by Blue Cross Blue Shield and Humana
- Medicare has the highest amount of transitions

That is all!

Thank you for your attention.

See you again soon!

Best Regards,
The Outliers Team

Data Cleaning: Abdulrahman Waheed

Data Modeling: Hoda Elkady

Data Curation: Mohammad Abdulhakim

Visualization: Abdulrahman Hisham